LABSHEET - 10

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ROLL NO. 03

BTECH CSE - B

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import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt
import numpy as np

# Load MNIST Dataset
(x_train, y_train), (x_test, y_test) = datasets.mnist.load_data()

# Normalize images
x_train = x_train / 255.0
x_test = x_test / 255.0

# Reshape for CNN
x_train = x_train.reshape((-1, 28, 28, 1))
x_test = x_test.reshape((-1, 28, 28, 1))

# CNN Model
model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu',
                  input_shape=(28,28,1)),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# Compile
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train
history = model.fit(
    x_train,
    y_train,
    epochs=5,
    validation_split=0.2
)

# Evaluate
test_loss, test_acc = model.evaluate(x_test, y_test)

print("Test Accuracy:", test_acc)

# Predict
prediction = model.predict(x_test[:5])

for i in range(5):
    print(
        f"Actual: {y_test[i]} | Predicted: {np.argmax(prediction[i])}"
    )
```

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Epoch 1/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 46s 29ms/step - accuracy: 0.9471 - loss: 0.1676 - val_accuracy: 0.9815 - val_loss: 0.0617
Epoch 2/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 45s 30ms/step - accuracy: 0.9838 - loss: 0.0527 - val_accuracy: 0.9859 - val_loss: 0.0498
Epoch 3/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 44s 29ms/step - accuracy: 0.9884 - loss: 0.0364 - val_accuracy: 0.9871 - val_loss: 0.0452
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Epoch 4/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 86s 32ms/step - accuracy: 0.9918 - loss: 0.0267 - val_accuracy: 0.9850 - val_loss: 0.0529
Epoch 5/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 49s 32ms/step - accuracy: 0.9934 - loss: 0.0208 - val_accuracy: 0.9880 - val_loss: 0.0445
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.9887 - loss: 0.0368
Test Accuracy: 0.9886999726295471
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 100ms/step
Actual: 7 | Predicted: 7
Actual: 2 | Predicted: 2
Actual: 1 | Predicted: 1
Actual: 0 | Predicted: 0
Actual: 4 | Predicted: 4
```

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Q1. What is image classification? How does it differ from object detection and image segmentation?

Answer: Image classification is the process of assigning a label or category to an image. Object detection identifies and locates objects using bounding boxes, while image segmentation classifies each pixel of the image into different categories.

Q2. Explain the architecture and working principle of a Convolutional Neural Network (CNN).

Answer: A CNN consists of convolutional layers, pooling layers, activation functions, fully connected layers, and an output layer. It learns image features automatically through convolution operations and uses these features for classification.

Q3. What is the role of Convolutional Layers, Pooling Layers, and Fully Connected Layers in a CNN?

Answer:

Convolutional Layers: Extract important features such as edges, shapes, and textures. Pooling Layers: Reduce feature-map size and computational complexity. Fully Connected Layers: Combine extracted features and perform final classification.

Q4. Why is image normalization performed before training a deep learning model?

Answer: Image normalization scales pixel values to a common range, which improves training speed, stability, and model accuracy.

Q5. Explain the purpose of activation functions such as ReLU and Softmax in CNNs.

Answer:

ReLU (Rectified Linear Unit): Introduces non-linearity and helps the network learn complex patterns. Softmax: Converts output values into probabilities and is used for multi-class classification.

Q6. What is a Confusion Matrix? How is it used to evaluate classification performance?

Answer: A Confusion Matrix is a table that compares actual labels with predicted labels. It helps evaluate classification performance by showing correct and incorrect predictions for each class.

Q7. Differentiate between Accuracy, Precision, Recall, and F1-Score with suitable examples.

Answer:

Accuracy: Percentage of total correct predictions. Precision: Percentage of correctly predicted positive cases out of all predicted positive cases. Recall: Percentage of correctly predicted positive cases out of all actual positive cases. F1-Score: Harmonic mean of Precision and Recall.

Q8. Compare traditional feature-based image classification methods (e.g., SIFT/HOG + SVM) with CNN-based classification.

Answer:

Traditional Methods (SIFT/HOG + SVM): Require manual feature extraction and separate classifiers. CNN-Based Methods: Automatically learn hierarchical features and generally provide higher accuracy and better performance.

Q9. Mention five real-world applications of CNN-based image classification.

Answer:

Handwritten digit recognition Facial recognition Biometric authentication Medical diagnosis Intelligent document processing

Q10. How can techniques such as Data Augmentation, Transfer Learning, and Hyperparameter Tuning improve the performance of deep learning-based image classification models?

Answer:

Data Augmentation: Increases dataset diversity and reduces overfitting. Transfer Learning: Uses pre-trained models to improve accuracy and reduce training time. Hyperparameter Tuning: Optimizes parameters such as learning rate, batch size, and epochs to achieve better model performance.